

LECTURE 9

GROUP AND RING THEORY

Lemma 6.22. *If R is a Noetherian ring, then each ideal contains a finite product of prime ideals.*

Proof. Let $\mathcal{F}$ be the family of ideals in R that do not contain any product of prime ideals. Suppose $\mathcal{F}$ is non-empty. Then by Theorem 6.13, the family $\mathcal{F}$ has a maximal element, say A . Since $A \in \mathcal{F}$, the element A is not a prime ideal. Hence there exist ideals B and C of R such that $BC \subseteq A$, but $B \not\subseteq A$ and $C \not\subseteq A$. Now

$$(B + A)(C + A) \subseteq BC + BA + AC + A^2 \subseteq A.$$

Since $B + A \supsetneq A$ and $C + A \supsetneq A$, both $B + A$ and $C + A$ contain a product of prime ideals, and hence $(B + A)(C + A)$ contains a product of prime ideals. This implies that A contains a product of prime ideals, a contradiction. Hence $\mathcal{F} = \emptyset$. $\square$

6.3. Hilbert basis theorem. We saw in Example 6.16 that $R[x]$ is not Artinian even if R is Artinian. This is not the case if R is Noetherian, as seen below.

Theorem 6.23 (Hilbert's basis theorem). *Let R be a Noetherian ring. Then the polynomial ring $R[x]$ is also a Noetherian ring.*

Proof. Suppose $I \subseteq R[x]$ is a left ideal that is not finitely generated. Then we can recursively construct a sequence of polynomials $\{f_0, f_1, \dots\}$ such that if J_n is the left ideal generated by $f_0, \dots, f_{n-1}$, then $f_n \in I \setminus J_n$ is of minimal degree. Without loss of generality set $J_1 = (0)$. By construction $\{\deg(f_0), \deg(f_1), \dots\}$ is a non-decreasing sequence of non-negative integers. Let a_n be the leading coefficient of f_n and let J be the left ideal in R generated by $\{a_0, a_1, \dots\}$. Since R is Noetherian, the chain of left ideals

$$(a_0) \subseteq (a_0, a_1) \subseteq \dots$$

terminates. Thus $J = (a_0, \dots, a_{N-1})$ for some non-negative integer N . In particular,

$$a_N = \sum_{i < N} u_i a_i, \quad \text{for some } u_i \in R.$$

Now consider

$$g = \sum_{i < N} u_i x^{\deg(f_N) - \deg(f_i)} f_i,$$

whose leading term is equal to that of f_N . Furthermore note that $g \in J_N$. However by construction $f_N \notin J_N$, and so if $\deg(f_N) > 0$, then $f_N - g \in I \setminus J_N$ having degree less than f_N yields a contradiction to the minimality of the degree of f_N . Finally if $\deg(f_N) = 0$, then instead $f_N - g = 0 \in I \setminus J_N$ is also a contradiction.

Here is an alternative proof:

Let $\mathcal{F}$ and $\mathcal{F}'$ be the families of left ideals of R and $R[x]$, respectively. Let $n \in \mathbb{Z}_{\geq 0}$ and define a mapping $\phi_n : \mathcal{F}' \rightarrow \mathcal{F}$, where

$$\phi_n(I) = \{a \in R \mid \exists ax^n + bx^{n-1} + \cdots \in I, a \neq 0\} \cup \{0\}.$$

It is easy to verify that $\phi_n(I) \in \mathcal{F}$. We claim that if $I, J \in \mathcal{F}'$ with $I \subseteq J$ and $\phi_n(I) = \phi_n(J)$ for all $n \geq 0$, then $I = J$. Let $0 \neq f(x) \in J$ be of degree m . Since $\phi_m(I) = \phi_m(J)$, there exists $g_m(x) \in I$ with leading coefficient the same as that of $f(x)$, and $f(x) - g_m(x)$ is either 0 or of degree at most $m - 1$. Suppose $f(x) - g_m(x) \neq 0$. As $f(x) - g_m(x) \in J$, we can similarly find $g_{m-1}(x) \in I$ such that $f(x) - g_m(x) - g_{m-1}(x) \in J$ and is either 0 or of degree at most $m - 2$. Continuing like this, we arrive, after at most m steps, at

$$f(x) - g_m(x) - g_{m-1}(x) - \cdots - g_i(x) = 0,$$

where $g_m(x), g_{m-1}(x), \dots \in I$. Then $f(x) \in I$, which proves that $I = J$, as claimed.

Let $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ be an ascending sequence of left ideals of $R[x]$. Then for each non-negative integer n ,

$$\phi_n(A_1) \subseteq \phi_n(A_2) \subseteq \phi_n(A_3) \subseteq \cdots$$

is an ascending sequence of left ideals of R . Hence there exists a positive integer $k(n)$ such that

$$(6.1) \quad \phi_n(A_{k(n)}) = \phi_n(A_{k(n)+1}) = \cdots.$$

Further, as R is Noetherian, the collection of left ideals $(\phi_n(A_i))$, for $n, i \in \mathbb{N}$, has a maximal element, say $\phi_p(A_q)$; compare Theorem 6.13. Then

$$\begin{aligned} \phi_p(A_q) &= \phi_n(A_q) && (\text{for all } n \geq p) \\ &= \phi_n(A_j) && (\text{for all } n \geq p, j \geq q). \end{aligned}$$

Therefore, we may choose $k(n) = q$ for all $n \geq p$ in (6.1). Moreover, if

$$s = \max\{k(1), \dots, k(p-1), q\},$$

then $\phi_n(A_s) = \phi_n(A_{s+1}) = \cdots$ for all $n \in \mathbb{N}$. Hence, by the result proved in the first paragraph, we obtain $A_s = A_{s+1} = \cdots$. Therefore $R[x]$ is Noetherian. $\square$

7. NIL IDEALS AND NILPOTENT IDEALS

In the last chapter, we focussed mainly on Noetherian rings towards the end. We now turn our focus a little more towards Artinian rings, and build up to a big result. First we need to introduce some definitions, which are important in their own right (also in the context of Noetherian rings), and some preliminary results.

Definition 7.1. Let R be a ring. An element $a \in R$ is *nilpotent* if there is an $n \in \mathbb{N}$ such that $a^n = 0$.

A left ideal I of R is a *nil (left) ideal* if every element in I is nilpotent.

A left ideal I of R is called *nilpotent* if there is an $n \in \mathbb{N}$ such that $I^n = 0$.

Note in the above that I^n denotes the set-wise product of n copies of I ; i.e.

$$I^n = \left\{ \sum_{k=1}^m i_{1,k} i_{2,k} \cdots i_{n,k} \mid m \in \mathbb{N}, i_{\ell,k} \in I \right\}.$$

Example 7.2. In the ring $\mathbb{Z}/9\mathbb{Z}$, the element 3 is nilpotent, since $3^2 \equiv 0 \pmod{9}$.

The matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

is nilpotent in $M_3(\mathbb{Z})$ as $A^3 = 0$.

Example 7.3. In the ring

$$R = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{C} \right\}$$

the ideal

$$I = \left\{ \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \mid b \in \mathbb{C} \right\}$$

is nil, and in fact, nilpotent. Indeed, we have $\begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, and also $I^2 = 0$ as

$$\begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & c \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

If I is nilpotent, then I is a nil ideal but not conversely, as seen below.

Example 7.4. Let $R = \bigoplus_{i \in \mathbb{N}} \mathbb{Z}/p^i\mathbb{Z}$ be the direct sum of $\mathbb{Z}/p^i\mathbb{Z}$ for a prime p . Then R contains non-zero nilpotent elements, such as

$$(0 + (p), p + (p^2), 0 + (p^3), \dots).$$

Let I be the set of all nilpotent elements. Then I is an ideal in R since R is commutative. So I is a nil ideal. However I is not nilpotent, for if $I^k = 0$ for some $k > 1$, then the element

$$x = (0 + (p), 0 + (p^2), \dots, 0 + (p^k), p + (p^{k+1}), 0 + (p^{k+2}), \dots).$$

is nilpotent, so it belongs to I . However $x^k \neq 0$, a contradiction. So I is not nilpotent.

If, however, the ring is Noetherian, the aforementioned converse holds. This is Levitzki's Theorem. However this theorem requires some addition material to state (such as what is a nilradical and the proof uses the concept of semiprime ideals), which is beyond the course content. We can however prove the converse more easily for just Artinian rings.

Theorem 7.5. *If J is a nil left ideal in an Artinian ring R , then J is nilpotent.*

Proof. Suppose for a contradiction that $J^k \neq 0$ for any $k \in \mathbb{N}$. Consider the family $(J, J^2, J^3, \dots)$. Since R is Artinian, this family has a minimal element (compare with Theorem 6.13), say $B = J^m$. Then $B^2 = J^{2m} \subseteq J^m = B$ implies $B^2 = B$. Consider another family

$$\mathcal{F} = \{A \mid A \text{ is a left ideal contained in } B \text{ with } BA \neq 0\}.$$

Then $\mathcal{F} \neq \emptyset$ because $B \in \mathcal{F}$. Let A be a minimal element in $\mathcal{F}$. Then $BA \neq 0$. This implies there exists an element $a \in A$ such that $Ba \neq 0$. However $Ba \subseteq A$ and $B(Ba) = B^2a = Ba \neq 0$. Thus $Ba \in \mathcal{F}$. Hence, by minimality of A , we have $Ba = A$. Thus there exists an element $b \in B$ such that $ba = a$. This implies $b^i a = a$ for all $i \in \mathbb{N}$. However, as b is a nilpotent element (since J is nil), this implies that $a = 0$, a contradiction. Therefore, for some $k \in \mathbb{N}$, we have $J^k = 0$. $\square$

Lemma 7.6. *Let R be a Noetherian ring. Then the sum of nilpotent (left) ideals in R is a nilpotent (left) ideal.*

Proof. Let $B = \sum_{i \in \Lambda} A_i$ be the sum of nilpotent ideals in R . Since R is Noetherian (i.e. left Noetherian), we have that B is finitely generated as a left ideal. Suppose $B = (x_1, \dots, x_m)$. Then each x_i lies in the sum of finitely many A_i 's. Hence B is contained in the sum of a finite number of A_i 's, say (after reindexing if necessary) $A_1, \dots, A_n$. Thus $B = A_1 + \dots + A_n$. Then by Exercise Sheet 10, we obtain that B is nilpotent. $\square$

Upon removing the condition of being Noetherian in the above, a counterexample exists: see Exercise Sheet 11. Note that it is not known whether the sum of two nil (left) ideals is still nil; this is Köthe's Conjecture.

Recall that if S is any non-empty subset of a ring R , then $\text{Ann}_R(S) = \{x \in R \mid xS = 0\}$ is a left ideal of R called the *left annihilator* of S in R . Recall also that an *ideal* means a two-sided ideal.

Theorem 7.7. *Let R be a Noetherian ring having no non-zero nilpotent ideals. Then R has no non-zero nil ideals.*

Proof. (Non-examinable for the oral exam) Suppose on the contrary that $N \neq 0$ is a nil ideal in R . Let

$$\mathcal{F} = \{\text{Ann}_R(n) \mid n \in N \setminus \{0\}\}$$

be a family of left annihilator ideals. Since R is Noetherian, by Theorem 6.13(iii) there exists $m \in N \setminus \{0\}$ such that $\text{Ann}_R(m)$ is a maximal element in $\mathcal{F}$.

Let $x \in R$. Then $mx \in N$, so there exists a smallest $k \in \mathbb{N}$ such that $(mx)^k = 0$. We claim that $mxm = 0$.

To be continued. $\square$